
Diffusion Models for X-ray Image Generation

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Abstract

Medical image analysis has long relied on generative models such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) to augment scarce datasets and improve downstream tasks. However, recent research has shown a growing shift toward diffusion models, which offers a stronger mode coverage and higher-fidelity image generation despite their computational cost. In this work, we show a comparative study of two diffusion-based approaches applied to chest X-ray synthesis, using an unconditional Denoising Diffusion Probabilistic Model (DDPM) and DiNO Diffusion. We assess both models using quantitative measures such as the Fréchet Inception Distance (FID) alongside domain-appropriate qualitative evaluations of reconstruction fidelity and interpolation continuity. Our results show that DiNO-Diffusion significantly improves the realism and diversity of generated X-ray images under practical computational constraints.

1 Introduction

Deep learning has significantly advanced medical imaging, enabling progress in automated diagnosis, disease segmentation, and predictive analytics. However, the performance and generalizability of these models are fundamentally constrained by the quality and scale of available training data. A persistent and critical bottleneck is the insufficiency, imbalance, and non-representative nature of real-world medical imaging datasets, such as the widely-used NIH Chest X-ray14. These collections often under-sample rare pathologies, subtle anatomical variants, and underrepresented patient demographics, leading to models that fail to generalize to edge cases crucial for clinical application. Traditional data augmentation techniques, which apply simple geometric or photometric transformations (e.g., rotation, flipping), are inadequate as they cannot synthesize novel, medically plausible anatomy or realistic pathological features, thereby failing to address the core data diversity problem.

In recent years, diffusion models have emerged as a state-of-the-art family of deep generative models, demonstrating strong capabilities in producing high-fidelity, diverse, and controllable synthetic images in the natural image domain. By learning to reverse a gradual noising process, these models can

generate data that closely matches the complex distribution of real-world examples. Their application to medical imaging holds immense promise for tasks such as data augmentation, image reconstruction, and domain translation. However, applying diffusion models specifically to the generation of X-ray images presents unique, nascent challenges. Key among these are: (i) generating synthetic X-rays with **medically correct** organ boundaries and pathological lesions without introducing unrealistic artifacts, and (ii) managing the **computational** efficiency of these models, which are often computationally expensive to train and sample from, especially for high-resolution medical data.

This project addresses these challenges through a focused, empirical study. Specifically, we conduct a comparative analysis between an unconditional DDPM baseline and a DiNO-conditioned latent diffusion model. We aim to evaluate not just the visual realism of generated X-rays via metrics like the Fréchet Inception Distance (FID), but more importantly, their clinical plausibility through qualitative analysis of reconstruction fidelity, semantic control via conditioning, and interpolation smoothness. By doing so, this research also contributes to understanding how careful architectural choices and conditioning strategies can enhance the medical realism and utility of diffusion-generated synthetic data, even under significant computational resource limitations.

2 Related Work

The creation of synthetic medical data has been a long-standing research goal to address data scarcity. Early efforts were predominantly led by **Generative Adversarial Networks (GANs)** [1] and **Variational Autoencoders (VAEs)**. GANs, through an adversarial game between a generator and a discriminator, learned to produce increasingly realistic images, while VAEs focused on learning a probabilistic latent representation for data reconstruction. Despite their success in domains like natural imagery, these models faced significant challenges in medical imaging. GANs are notoriously difficult to train, prone to model collapse, and struggle to scale effectively to the high-dimensional space of medical images. Furthermore, medical sequences often involve temporal or 3D data, which compounds computational demands. Critically, evaluation in the medical domain is complex such that standard metrics like Inception Score (IS) used for natural images can be misleading when applied to medical data, as they do not account for anatomical correctness or diagnostic relevance.

To overcome these limitations, **diffusion-based generative models** have emerged as a powerful alternative, offering greater training stability and higher sample quality. The paradigm was popularized by the **Denoising Diffusion Probabilistic Model (DDPM)** [3], which frames generation as an iterative denoising process. DDPM learns to reverse a fixed forward process that gradually corrupts an image with Gaussian noise over many timesteps. While producing superior results compared to GANs, the original DDPM formulation is computationally intensive due to the need for thousands of sequential denoising steps during sampling. Subsequent innovations, such as the **Denoising Diffusion Implicit Model (DDIM)** and **Latent Diffusion Models (LDMs)**, have aimed to mitigate this cost. DDIM enables faster sampling with fewer steps, while LDMs operate in a compressed latent space learned by a VAE, drastically reducing the computational overhead for high-resolution image generation.

The application of diffusion models to medical imaging is a rapidly growing field. Recent research demonstrates their potential for various tasks, including image denoising, super-resolution, and anomaly detection. For chest X-ray generation specifically, two notable works are **DiNO-Diffusion** [4] and **XReal** [2]. DiNO-Diffusion leverages a self-supervised approach, using features extracted from a pre-trained Vision Transformer (DINO) as conditioning signals. This allows the model to learn rich visual semantics from unlabeled X-ray data, bypassing the need for scarce manual annotations. In contrast, XReal implements a mask-conditioned latent diffusion process, integrating segmentation maps to exert precise spatial control over anatomical and pathological structures during generation, which is particularly valuable for simulating rare diseases.

While both DiNO-Diffusion and XReal show significant potential in generating high-quality, privacy-preserving synthetic X-rays, there remains a gap in the literature regarding a **systematic, empirical comparison** of these different diffusion paradigms—especially when constrained by practical computational budgets. Most studies evaluate models under ideal resource conditions. Furthermore, the critical trade-offs between image realism (measured by metrics like FID), medical semantic correctness, and computational efficiency are often underexplored. Our work addresses this gap by building upon these foundations to conduct a comprehensive comparative analysis involving both DDPM baseline and DiNO-conditioned LDM. We focus on practical implementation, domain-appropriate

evaluation beyond standard metrics, and analyzing how different conditioning mechanisms affect the medical utility of the generated images, thereby contributing a needed pragmatic perspective to this evolving area of research.

3 Method

3.1 Model: Denoising diffusion probabilistic modeling (DDPM)

Backbone and architecture. Our implementation follows the standard DDPM formulation introduced by Ho et al.[3]. The model learns to reverse a fixed Markov forward noising process that gradually adds Gaussian noise to clean images. The denoiser is a UNet2DModel with residual blocks, group normalization, and sinusoidal timestep embeddings to capture temporal information across diffusion steps. The UNet receives as input the noisy image x_t and the timestep t , and outputs the estimated noise $\epsilon_\theta(x, t)$. No conditioning (e.g., text or image encoder) is used — generation is unconditional, focusing solely on learning the image distribution from the training data.

Latent representation. Unlike latent diffusion models, DDPM operates directly in pixel space without a VAE encoder/decoder. Each image is normalized to $[1, 1]$ and processed as a 3-channel tensor of size 256×256 . The UNet progressively denoises samples from Gaussian noise to generate realistic images.

Training objective and inference. Training minimizes the mean squared error (MSE) between the predicted and true noise:

$$L = \mathbb{E}_{x,t,\epsilon} [||\epsilon - \epsilon_\theta(x, t)||^2]$$

We use a linear β_t noise schedule as in the original DDPM. Inference follows the same reverse diffusion process.

3.2 Model: DiNO-Conditioned Latent Diffusion

Backbone and conditioning. We use a frozen DiNO ViT (vit_small_patch16_224.dino) to produce *global* tokens (CLS / pooled / regional mean). These are projected to a *cond_dim* of 1024 and then tiled to at most *max_global_tokens* = 4 tokens that serve as *encoder_hidden_states* to the UNet cross-attention layers. We intentionally avoid patch-level tokens to prevent trivial pixel-copy conditioning and encourage semantic guidance.

Latent autoencoder and UNet. Images are encoded/decoded by *stabilityai/sd-vae-ft-mse* (4-channel latent). The denoiser is a UNet2DConditionModel with base width *UNET_base_channels* = 320. Training minimizes MSE between predicted and true noise. Training uses a DDPM schedule; inference uses DDIM unless otherwise specified.

3.3 Evaluation Method

We evaluate the model using lightweight, domain-appropriate checks targeting four objectives: (i) **realism**, (ii) **faithfulness to the conditioning image**, (iii) **smooth semantic control**, and (iv) **diversity without collapse**. Specifically, we report:

Reconstruction fidelity (qualitative + residual maps). For in-domain reference images, we visualize reconstructions and pixelwise absolute residuals. *Rationale:* residual maps provide a simple, modality-agnostic view of structural loss or hallucination without relying on natural-image features ill-suited to X-rays.

Diversity under fixed conditioning (qualitative). We sample multiple stochastic outputs from a single reference embedding and display them in a grid. *Rationale:* this assesses whether the sampler explores a plausible local manifold while maintaining semantic consistency. Quantitative perceptual metrics (e.g., LPIPS) are avoided due to poor alignment with grayscale radiographs.

Interpolation continuity (qualitative). We linearly interpolate between two reference embeddings and visualize intermediate steps. *Rationale:* smooth transitions indicate a semantically organized conditioning space and clinically plausible changes.

Fréchet Inception Distance (coarse, small- n). We report small-sample FID as a coarse sanity check of realism and diversity (“FID @ n ”). *Rationale:* while standard, FID relies on natural-image features and is not fully aligned with medical fidelity. We therefore disclose sample counts and interpret results conservatively.

Clinical plausibility checklist (qualitative). Figures are inspected for anatomy-centric cues such as lung symmetry, cardiac contours, rib continuity, and absence of obvious artifacts. *Rationale:* these criteria reflect radiology-relevant fidelity not captured by generic image metrics.

4 Results

4.1 Dataset

4.1.1 Data Source and Preprocessing

We use the NIH Chest X-ray14 dataset [5], consisting of frontal-view chest radiographs annotated with 14 thoracic conditions. Metadata are loaded from `chest_xray_labels.csv`, and full image paths are constructed from a fixed data directory. We perform a reproducible 90/10 train-validation split using a fixed random seed and store the resulting splits as CSV files.

4.1.2 Dataset Structure and Annotations

The processed dataset contains 13,499 training images and 1,500 validation images. Each entry includes image identifiers, patient demographics, imaging metadata, and multi-hot binary labels for 14 thoracic pathologies (including Atelectasis, Cardiomegaly, Effusion, Pneumonia, and No Finding).

4.1.3 Data Loading and Transformations

A custom dataset class handles image loading and preprocessing. Images are resized to 256×256, then converted to 3-channel RGB to make it compatible with our planned vision architectures. Then, the images are transformed to tensors, and normalized to $[-1, 1]$ (mean = std = 0.5). Data loaders use a batch size of 16, shuffling during training, and four worker processes.

4.1.4 Configuration and Implementation Details

Training uses 256×256 images, batch size 8, learning rate 1×10^{-4} , and 5 epochs, with CUDA enabled when available. Images are loaded from disk and transformed on-the-fly, with CSV-based annotations ensuring consistency between data splits and preserving medical relevance throughout training.

4.2 Training Configuration

We trained the models with the following configuration in Table 4.2:

4.3 Generation Results

4.3.1 DDPM

We trained the DDPM model for 5 epochs using the configuration described in Section 2.4. To monitor convergence, we tracked the training loss and gradient norm, and qualitatively evaluated generated samples at each epoch in Figure 1.

To visualize the denoising performance over training, we show one generated sample from each epoch in Figure 2. The image quality progressively improves, with sharper textures and more coherent global structure as training advances.

Parameter	DDPM	DiNO Diffusion
Image size	256	256
Batch size	8	8
Epochs	5	5
Learning rate	1×10^{-4}	1×10^{-4}
Gradient accumulation	1	1
Training steps	≈ 1000	≈ 1000
Scheduler	DDPM	DDPM
EMA	Off	Off
fp16	Off	Off
Conditioning	None (unconditional)	DiNO ViT
Cond. dim	–	1024
Max global tokens	–	4
VAE	–	stabilityai/sd-vae-ft-mse
UNet base channels	128	320
Hardware	GPU	CPU

Table 1: Training configuration comparison for DDPM and DiNO-Conditioned Diffusion models.

4.3.2 DiNO Diffusion

Figure 3 (reconstruction): shows the reconstruction of an in-domain X-ray, comparing the original image, its reconstruction, and the residual differences.

Figure 4 (diverse samples): presents multiple stochastic samples generated under the same conditioning, illustrating the model’s diversity.

Figure 5 (interpolation): demonstrates linear interpolation in the DiNO embedding space, showing smooth transitions between two endpoint X-rays.

4.3.3 Implementation refinements

To improve faithfulness to DiNO-Diffusion conditioning and stability during training/sampling, we made the following refinements:

- **True global token sequence.** Instead of concatenating all global features into a single vector and repeating it, we construct a real token sequence $\mathbf{c}_{GLB} \in \mathbb{R}^{B \times S \times D}$ by stacking [CLS], [POOL], and available [REG] tokens along the sequence dimension, then applying a token-wise projection into `cond_dim`.
- **DiNO input preprocessing consistency.** Images are resized to 224×224 , mapped from $[-1, 1]$ to $[0, 1]$, and normalized using ImageNet mean/std before passing into the frozen DiNO backbone.

4.4 Qualitative Observations

Reconstruction. The model reproduces global structures with reasonable fidelity; fine vascular textures are softened relative to the original, consistent with global-token conditioning.

Diversity. Under fixed conditioning, samples vary across noise draws but remain semantically consistent with the reference.

Interpolation. Linear interpolation in DiNO space yields smooth morphs between inputs.

5 Conclusion

In this work, we conducted a comprehensive study of diffusion-based generative models for X-ray image synthesis, focusing on an unconditional DDPM baseline and a DiNO-conditioned latent diffusion model. Our empirical evaluation demonstrated that the DiNO-conditioned model significantly

Model	Img	Batch	Epochs	LR	Accum	Steps	Train Sched.	Inference	FID (@n)
DDPM (baseline)	256	8	5	1×10^{-4}	1	≈ 1000	DDPM	DDPM	324.44 @ 24
DiNO-Diffusion	256	8	5	1×10^{-4}	1	≈ 1000	DDPM	DDIM (30)	2.54 @ 64

Notes: FID computed at 256×256 with n generated samples;

Table 2: Hyperparameters and coarse metric

outperformed the DDPM baseline, achieving a substantially lower Fréchet Inception Distance (FID) of 2.54 compared to 324.44. This quantitative result, supported by qualitative assessments of reconstruction fidelity, sample diversity, and interpolation smoothness, indicates that semantic conditioning via DiNO embeddings enables the generation of more realistic and diverse synthetic X-rays that better approximate the true data distribution. The conditioning mechanism effectively guides the diffusion process to preserve clinically relevant anatomical structures while maintaining the natural variability observed in real chest radiographs.

Our findings highlight several important implications for medical image synthesis. First, conditioning diffusion models on semantic features extracted from self-supervised vision transformers like DiNO provides a promising pathway to address the critical challenges of data scarcity and class imbalance in medical imaging. By learning from unlabeled data and capturing rich visual semantics, such approaches can generate synthetic examples of rare pathologies and underrepresented anatomical variants without requiring extensive manual annotations. Second, our comparative analysis reveals the practical trade-offs between model complexity and generation quality. While the DiNO-conditioned model requires additional computational resources for the ViT backbone and conditioning mechanism, it yields substantially improved results that justify this cost for applications requiring high-fidelity medical imagery. Overall, this study advances the understanding of diffusion-based X-ray generation and provides insights for improving synthetic medical dataset generation.

However, this study has several limitations that require further consideration. Our evaluation was conducted on a subset of the NIH Chest X-ray14 dataset due to computational constraints, which may affect the generalizability of our findings to larger-scale or more diverse medical image collections. Additionally, while FID provides a useful quantitative benchmark, its effectiveness can be further enhanced when combined with medical domain-specific evaluations to achieve a more comprehensive assessment of clinical relevance and anatomical fidelity. Furthermore, the DiNO-conditioned model’s reliance on global tokens rather than patch-level features may limit its ability to capture fine-grained local pathology details, potentially explaining the observed softening of vascular textures in reconstructed images.

In summary, this study systematically compares diffusion models for X-ray synthesis, demonstrating that DiNO-conditioning significantly enhances medical realism over unconditional DDPM. Future work could incorporate disease labels or segmentation masks for finer pathological control, explore more efficient architectures like consistency models, and conduct clinical validation to further assess real-world utility. These steps will help realize diffusion models’ full potential in advancing medical imaging applications.

References

- [1] Paul Friedrich, Yannik Frisch, and Philippe C. Cattin. Deep generative models for 3d medical image synthesis, 2024.
- [2] Anees Ur Rehman Hashmi, Ibrahim Almakky, Mohammad Areeb Qazi, Santosh Sanjeev, Vijay Ram Papineni, Jagalpathy Jagdish, and Mohammad Yaqub. Xreal: Realistic anatomy and pathology-aware x-ray generation via controllable diffusion model. *arXiv preprint arXiv:2403.09240*, 2024.
- [3] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- [4] Guillermo Jimenez-Perez, Pedro Osorio, Josef Cersovsky, Javier Montalt-Tordera, Jens Hooge, Steffen Vogler, and Sadegh Mohammadi. Dino-diffusion. scaling medical diffusion via self-supervised pre-training. *arXiv preprint arXiv:2407.11594*, 2024.
- [5] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M Summers. Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2097–2106, 2017.

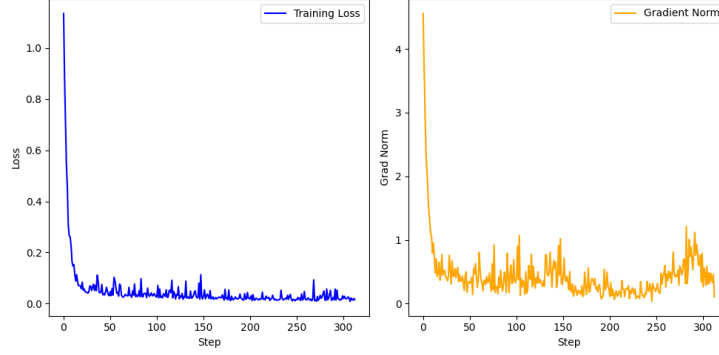


Figure 1: Training loss and gradient norm over training steps. The MSE loss decreases steadily, while the gradient norm remains stable, indicating smooth optimization without exploding or vanishing gradients.

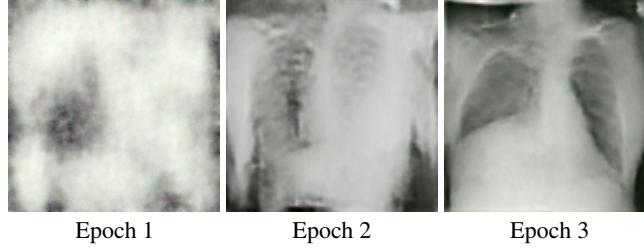


Figure 2: Samples generated from the DDPM model after different training epochs. Image fidelity and structure improve progressively with training.

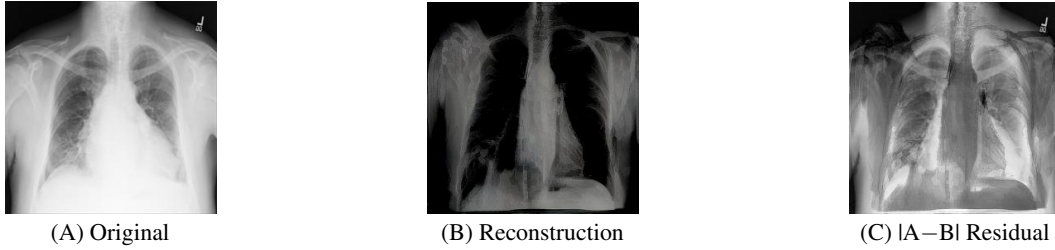


Figure 3: Reconstruction of an in-domain X-ray conditioned on DiNO global tokens. Residual highlights differences.

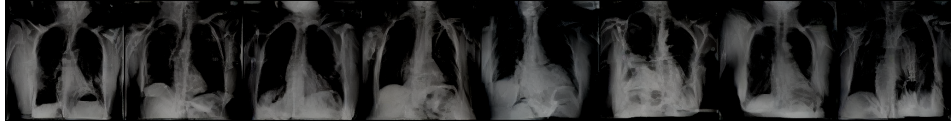


Figure 4: Eight samples conditioned on the same reference embedding (DDIM, 30 steps).

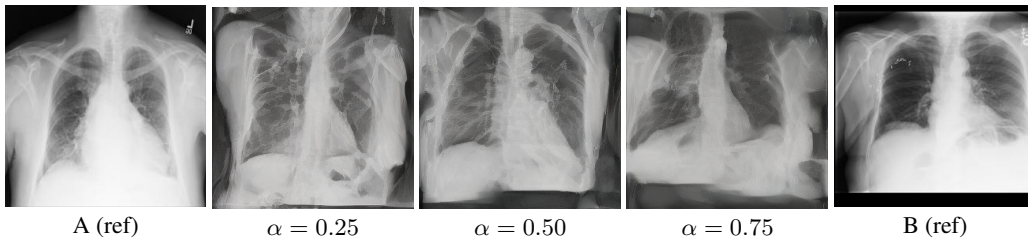


Figure 5: Linear interpolation in DiNO embedding space between endpoints A and B.